

Cooper Union School of Engineering
Principles of Probability Syllabus
Math 224 - Section A
Fall 2026

1. COURSE INFORMATION

Instructor: Dr. Abigail Raz

You may refer to me as Professor Raz or Dr. Raz. (Please do not use anything else.)

Email: abigail.raz@cooper.edu (This is my preferred mode of contact. You may also message me on Teams but I will likely take longer to respond.)

Website: There is a course site on Teams. This is where I will post announcements and course materials. It is your responsibility to check Teams regularly and ensure that you are receiving the emails.

Office: 41 Cooper Square Room 313.

Office Hours: (Subject to change) Unless otherwise stated all office hours are held in my office. My office hours are drop-in; no need to make an appointment during these times, just come on by!

Monday 10:15am-11:45am

Tuesday 3pm-5pm

Thursday 1pm-3pm

The current plan is to hold all classes and office hours in person. However, if that changes during the semester please use the following link.

Zoom Meeting ID: 950 7945 9992

Zoom Passcode: 41CS313.

Additional office hours are available by appointment. If you wish to meet with me you must email me no later than 5 pm on the day before you would like to meet. I will do my best to accommodate your schedule, but **please send multiple times that you are available to meet for me to choose from.**

Prerequisites: MA223/MA225 as a co-requisite

Credits: 2

Class Time and Location:

Monday 2pm-2:50pm in Room 105

Wednesday 2pm-2:50pm in Room 504

Textbook: Our primary textbook is listed below.

- Hogg, Robert V. and Tanis, Elliot A. *Probability and Statistical Inference* Eighth Edition. Pearson Prentice Hall, 2010. Digital Library Copy

I will have a copy of the text in my office if you need to reference anything. There is also a physical copy on course reserve in the library.

Course Description and Objectives:

Random events have fascinated people for centuries. In this class we will learn about the rigorous mathematical language describing random events and their probabilities (these concepts began with Kolmogorov's study of Probability Theory in the 1930's). The goal of this course is for you to become familiar with this probabilistic language and how to solve the associated problems. We will cover many concepts throughout the course and discuss how they tie together. Most importantly, this course will provide a foundation for any further interactions with probability whether in daily life or in additional study.

The methods and approaches used in a probability course can often feel quite different to other math courses, such as calculus, which you have taken. In particular, it often becomes shockingly simple to arrive at wildly implausible answers. Please make sure you are consistently taking the time to reflect on your work and not only consider if the individual steps are correct, but, additionally, if the final answer is plausible given the framework of the question. I strongly recommend reading assignments as soon as possible and allowing yourself the time to be confused, try ideas, fail, work with your classmates and myself, and then eventually arrive at a complete solution. If you find yourself falling behind with the material please come to office hours or schedule a meeting as soon as possible.

I will periodically break up the class into groups to work on problems. This group work will not be graded, but I reserve the right to change that if focus during this period becomes an issue. Additionally, I will occasionally ask for a student to present a solution to the group work on the board.

Cooper Union Policies:

Cooper Union strives to make all learning experiences as accessible as possible. If you anticipate or experience barriers based on your disability (including mental health, chronic or temporary medical conditions), please let me know immediately so that we can discuss options privately. If you are eligible for services and register with the office of student affairs, please arrange a meeting with me as soon as possible to discuss your accommodations so they can be implemented in a timely manner. More information can be found at <https://cooper.edu/students/student-affairs/disability/students>. Cooper Union also provides a variety of counseling and mental health services. More information can be found at <https://cooper.edu/students/student-affairs/health/counseling>.

Additionally, while I want you to feel comfortable coming to me, please be aware as a member of the faculty I have certain reporting requirements. If you inform me of an issue of sexual harassment, sexual assault, or discrimination, I will keep the information as private as I can. However, I am required to report the basic facts of the incident to Cooper's Title IX Coordinator. The Cooper Union Title IX policy on sexual misconduct can be found [here](#).

All students in the course are expected to be familiar with and abide by the academic integrity policy (<https://cooper.edu/engineering/curriculum/academic-standards-regulations>). Violations of the policy are taken very seriously.

2. MISSING CLASS

While I will not take attendance each day it is important for your understanding of the course material to miss as little class as possible. However, I understand that for most students there will be at least one class that you will need to miss for a variety of reasons. If you do miss a class please follow the guidelines below:

- (1) As soon as you know you will be missing class ask a classmate for a copy of their notes from that day.
- (2) If a written homework is due during class please email it to me as a pdf no later than the end of the class period.
- (3) If you have any questions on the material from the class notes or simply want to double check there is nothing additional that you have missed I strongly encourage you to attend my office hours or schedule an individual meeting.
- (4) You do not need to provide any documentation related to your absence.
- (5) Make-up exams will only be given in genuine and unavoidable circumstances. You should contact me as soon as you know you will not be able to attend the exam and we will discuss an alternative plan.
- (6) **Regarding exams and religious observance: If there is a conflict with your religious observance please let me know ASAP and we will arrange another time for you to take the exam. For all alternate scheduling the earlier you let me know the better, and you must let me know at least two weeks before the relevant exam. Please note that this includes conflicts due to work restrictions, specific religious services, as well as afternoon exams on a fast day.**

3. ASSIGNMENTS AND GRADES

Participation	4%
Combined Homework	16%
Quizzes (x3)	4% each (12% total)
Midterms (x2)	16% each (32% total)
Final	36%

Final letter grade: At the end of the course your final grade will be based on your percentage. I do not decide on final grade cutoffs until the end of the semester, but it will not be any worse than the following table:

$\leq \text{Percentage} <$	90-100	80-90	70-80	60-70	< 60
Grade	A	B	C	D	F

Participation: Every student will start with a base participation grade of 90%. You can gain points by being attentive, asking or answering questions in class, and attending office hours. You will lose points here for poor attendance or distracting behavior such as consistently checking your phone during class. I expect interactions in class to be civil, respectful, and supportive of an inclusive learning environment for all. These should be easy points.

Homework: There will be written homework, due in class, almost every week (please see the schedule at the end of the syllabus for the exact dates). The homework is split into three parts.

Part 0 contains suggested problems out of our textbook that will not be turned in or graded. Often they will be similar to questions on other parts of the homework. I encourage you to look through

these problems and do as many of them as you need to become comfortable with the material.

Part 1 will be graded by a grader, while I will grade part 2. They must be turned in separately since they will go to two different people. Homework must be legible, stapled, and have ample space for comments. Late homework will not be accepted; if you know you cannot make class you may email me your homework. The assignment must be emailed as a pdf and in my mailbox no later than the end of that days class. While there will be a total of 12 homework assignments the homework grade will be computed as follows. Let P be the total number of homework points assigned. Let $Q = \frac{11P}{12}$. Your homework percentage will be:

$$\frac{\min\{Q, \text{ the total number of homework points you have earned}\}}{Q}.$$

While you are allowed and encouraged to work with your classmates on the homework the write-up that you ultimately hand in should be completely your own. At no point should you copy a solution from another student or any other source. At the top of every homework please list the students with whom you worked on that assignment.

Quizzes: There will be three quizzes; see the calendar below for the dates. Make-ups will only be given in genuine and unavoidable circumstances. You should contact me as soon as you know you will not be able to attend the quiz. Each quiz will be comprised of three True/False questions. You must both decided if the statement is true or false and give a complete justification. There will be not be any method for corrections on quizzes.

Midterms: There will be two midterms; see the calendar below for the dates. Make-ups will only be given in genuine and unavoidable circumstances. You should contact me as soon as you know you will not be able to attend the exam.

You will be allowed to earn back points on the midterms in two different ways, and on any midterm you may use a combination of the two correction types.

You may only earn up to 80% of the total exam score after corrections.

- (1) **Written Corrections:** You may submit a written correction of any midterm problem up to a week after you receive the graded exam. You can earn up to 50% back of the points you lost on that problem (i.e. if you receive a 80% on a problem you can submit a written correction and receive up to 90% on that problem). You must turn in the following:
 - The original graded exam. Please do not make any additional markings directly on the exam.
 - A complete solution of the problem.
 - An explanation of why you lost points or were unable to start the problem as well as an outline of the tools you ultimately did use to solve it. Depending on the problem this could be as short as a sentence or two or as long as two paragraphs.
- (2) **Oral Corrections:** If you earn below a 50% on the exam then you may meet with me for an oral retake of any problem where you earned less than 30%. You can earn up to 65% of the total points on each corrected problem. Please email me within 24 hours of getting your exam back to set up a time to take this oral retake. We will discuss the original problems and I may also give you another similar problem to work through with me.

Note that this policy only applies to the midterms and not to quizzes or the final exam.

Final Exam: As the final exam week at Cooper obeys the same schedule as a standard week of classes we will be having two final exams - one during each class.

4. CALENDAR

Important University dates:

- September 8: Add/drop deadline.
- October 27: Last day to withdraw from classes.

Below is the tentative calendar of topics and assignments for the semester.

Week	Material and Reading	Monday	Wednesday
Aug 30	Introduction to and properties of probability 1.1, 1.2		
Sep 7	Counting! 1.3	No Class Labor Day HW 0 (Tuesday)	HW 1
Sep 14	More counting and conditional probability 1.3-1.4		HW 2
Sep 21	Conditional probability and independence 1.4-1.5	No Class Video lecture posted on Sep 17	HW 3
Sep 28	Bayes' theorem and discrete random variables 1.5-1.6,2.1		HW 4
Oct 5	More on discrete random variables and expected value 2.1-2.2	Quiz 1	HW 5
Oct 12	Expected value and variance 2.2-2.3		HW6
Oct 19	Binomial distribution 2.4	Exam 1	
Oct 26	Moment generating functions and Poisson distribution 2.5-2.6		HW 7
Nov 2	Continuous type date 3.1-3.2	Quiz 2	HW 8
Nov 9	Continuous random variables 3.3		HW 9
Nov 16	Expectation and variance of continuous random variables, uniform and exponential distributions 3.3-3.4		
Nov 23	None	Exam 2 HW 10	No class Friday classes held
Nov 30	Normal Distribution and functions of random variable 3.6, 5.1		HW 11
Dec 7	Functions of a random variable and survey of tail bounds and limit theorems (time permitting) 5.1		Quiz 3 HW 12
Dec 14	Final exam week	Final Exam Part 1	Final Exam Part 2